

right? From this point of view, the mean and variance of arithmetic returns are useful statistical measures.

Here's another example where using the arithmetic mean gets it right and using geometric means is wrong. Suppose that asset A and asset B have average gross returns of $E[R_{t+1}^A] = 1.5$ and $E[R_{t+1}^B] = 0.5$, respectively. An investor holding 50% of asset A and 50% of asset B breaks even, on average, with the arithmetic mean being zero. Taking the average of asset A and B's geometric returns would actually result in a number less than zero. Log returns do not aggregate across portfolios. In both examples, however, we are better off looking at the full distribution of returns.